

HEART FAILURE

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HEART FAILURE MANAGEMENT TIPS

1. Assess functional status (NYHA class)
2. Assess volume status (symptoms and signs, including weight)
3. Assess adherence to medications and fluid restriction
4. Medication review to ensure patients are on appropriate disease modifying treatment
5. Control cardiovascular risk factors
6. Assess renal function and electrolytes regularly
7. Advise on exercise, educate patients and screen for psychosocial issues

Heart Failure with Reduced Ejection Fraction (HFrEF) (LVEF $\leq$ 40%)

Ensure 4 pillars of HFrEF have been initiated:

- ✓ *ARNI OR ACEI / ARB
- ✓ Beta blockers (bisoprolol, carvedilol)
- ✓ MRA
- ✓ SGLT-2 inhibitors

Heart Failure with Mildly Reduced and Preserved Ejection Fraction (Non HFrEF) (LVEF $>$ 40%)

- Initiate SGLT-2 inhibitors
- Consider other HFrEF treatment for suitable patients

Red Flags for ED Referral / Early Cardio Review

- ▶ ADHF with pulmonary oedema
- ▶ Hypotension (SBP $<$ 80mmHg)
- ▶ Rapid worsening of renal function
- ▶ Fluid overload not responding to trial of increased diuretics
- ▶ First presentation of heart failure

**Preferred option but recommended for 2DE to review EF prior to initiation. Usually started by Cardiologists.*

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INTRODUCTION

Clinical Definition

Heart failure (HF) is a clinical syndrome with symptoms and/or signs caused by a structural and/or functional cardiac abnormality and corroborated by elevated natriuretic peptide levels and/or objective evidence of pulmonary or systemic congestion (*Universal Definition and Classification of Heart Failure 2021*).

Epidemiology

4.5% of the Singapore population was found to be living with heart failure. Nearly 10% of people in their 80s suffer from heart failure.

DIAGNOSIS, CLASSIFICATION, AETIOLOGY AND SEVERITY OF HEART FAILURE

Diagnosis of Heart Failure (Universal Definition and Classification of Heart Failure 2021)

Criteria 1 must be fulfilled and corroborated by either criterion 2 or 3.

1. Symptoms and/or signs of heart failure caused by a structural and/or functional cardiac abnormality.
2. Ejection fraction (EF) < 50%, abnormal cardiac chamber enlargement, elevated left ventricle filling pressures (i.e., E/E' of > 15), moderate / severe ventricular hypertrophy or moderate / severe valvular obstructive or regurgitant lesion.
3. Elevated natriuretic peptide levels.
4. Objective evidence of cardiogenic pulmonary or systemic congestion by diagnostic modalities (e.g., chest radiograph or elevated filling pressures by echocardiography) or haemodynamic measurement (e.g., right heart catheterisation, pulmonary artery catheterisation) at rest or with provocation.

Classification of Heart Failure

1. Heart Failure with reduced ejection fraction (HFrEF): HF with EF $\leq$ 40%
2. Heart Failure with mildly reduced ejection fraction (HFmEF): HF with EF 41 to 49%
3. Heart Failure with preserved ejection fraction (HFpEF): HF with EF $\geq$ 50%
4. Heart Failure with improved EF (HFimpEF): HF with a baseline EF of $\leq$ 40%, a $\geq$ 10-point increase from baseline EF, and a second measurement of EF of > 40%

Initial Assessment

Symptoms: Ask for Chest Pain, Syncope and Palpitations Routinely

1. Breathlessness, e.g., orthopnoea, PND
2. Pedal oedema
3. Fatigue, e.g., dyspnoea at rest, dyspnoea on physical activity

Signs

1. Weight gain
2. Raised jugular venous pressure
3. Displaced apical beat
4. Peripheral oedema
5. Lung crepitations
6. Third heart sound
7. Tachycardia

Symptoms	Signs
Typical	More specific
Breathlessness Orthopnoea Paroxysmal nocturnal dyspnoea Reduced exercise tolerance Fatigue, tiredness, increased time to recover after exercise Ankle swelling	Elevated jugular venous pressure Hepatojugular reflux Third heart sound (gallop rhythm) Laterally displaced apical impulse
Less typical	Less specific
Nocturnal cough Wheezing Bloated feeling Loss of appetite Confusion (especially in the elderly) Depression Palpitation Dizziness Syncope Bendopnea ^a	Weight gain (>2 kg/week) Weight loss (in advanced HF) Tissue wasting (cachexia) Cardiac murmur Peripheral oedema (ankle, sacral, scrotal) Pulmonary crepitations Pleural effusion Tachycardia Irregular pulse Tachypnoea Cheyne-Stokes respiration Hepatomegaly Ascites Cold extremities Oliguria Narrow pulse pressure

Taken from ESC Heart failure guidelines (2021)

Aetiology of Heart Failure

1. Coronary artery disease
2. Hypertension
3. Valvular heart disease
4. Cardiomyopathy (familial and nonfamilial)
5. Thyroid disease
6. Diabetes mellitus
7. Anaemia
8. Post-myocarditis
9. Previous cancer treatment (chemotherapy induced cardiomyopathy, radiation heart disease)
10. Infiltrative diseases (amyloidosis, haemachromatosis, glycogen storage diseases)
11. Toxins (alcohol, cocaine)
12. Associated with neuromuscular diseases

Precipitating Causes of Heart Failure

A. Non-Cardiac

1. Non-compliance to medications, fluid / salt restriction
2. Concomitant medications, e.g., NSAIDS, calcium channel blockers (except amlodipine and felodipine), thiazolidinediones, e.g., rosiglitazones, metformin (does not precipitate heart failure but not to be used in renal impairment, as may cause lactic acidosis) (in renal impairment with high creatinine), anti-arrhythmics other than amiodarone
3. Alcohol abuse
4. Renal dysfunction
5. Infection, e.g., UTI, pneumonia, sepsis
6. Pulmonary embolism
7. Thyroid dysfunction
8. Anaemia (occult bleed)
9. Suboptimal treatment, e.g., not on angiotensin-converting enzyme (ACE) inhibitors and diuretics

B. Cardiac

1. Arrhythmias: atrial fibrillation, supraventricular or ventricular arrhythmias
2. Myocardial ischaemia
3. Valve leaflet dysfunction, secondary to papillary muscle rupture

Complications of Heart Failure

1. Arrhythmias, e.g., AF, VT, VF bradyarrhythmias
2. Thromboembolism, e.g., stroke, DVT, pulmonary embolism
3. Gastrointestinal, e.g., hepatic congestion / dysfunction, malabsorption
4. Musculoskeletal, e.g., muscle wasting, cachexia
5. Respiratory, e.g., pulmonary congestion / hypertension, respiratory muscle weakness

Severity of Heart Failure

New York Heart Association (NYHA) Classification

Class	Severity	Symptoms
I	Asymptomatic	No symptom with ordinary physical activity
II	Mild	Comfortable at rest but ordinary activity causes symptoms
III	Moderate	Comfortable at rest but symptoms with less than ordinary activity
IV	Severe	Symptomatic at rest and without any physical activity

Stages of Heart Failure (AHA/ACC)

Stages	Definition
A	At risk for HF but without current or prior symptoms or signs of HF and without structural or biomarkers evidence of heart disease
B	Structural heart disease or abnormal cardiac function, or elevated natriuretic peptide levels without current or prior symptoms or signs of HF
C	Current or prior symptoms and/or signs of HF caused by a structural and/or functional cardiac abnormality
D	Severe symptoms and/or signs of HF at rest, recurrent hospitalisations despite guideline-directed management and therapy (GDMT), refractory or intolerant to GDMT, requiring advanced therapies such as consideration for transplant, mechanical circulatory support, or palliative care.

INVESTIGATIONS AT PRIMARY CARE

A. Non-Blood Investigations

1. Chest X-ray

- Useful mnemonic of ABCDE
 - A - alveolar oedema (bat wing opacities)
 - B - Kerley B lines
 - C - cardiomegaly
 - D - dilated upper lobe vessels
 - E - pleural effusion
- May also detect conditions causing shortness of breath and mimicking features of heart failure.

2. ECG

- Patients with chronic heart failure are unlikely to present with a normal electrocardiogram; however, 20% may have a normal electrocardiogram. Commonly seen ECG abnormalities include left ventricular hypertrophy, left axis deviation, left bundle branch block, and pathological Q-waves.
- Arrhythmias e.g., sinus tachycardia, atrial fibrillation and acute S-T segment changes may be seen in cases of decompensated heart failure.

3. Urinalysis

- Screen for proteinuria and glycosuria

B. Blood Investigations

1. Full blood count

- Anaemia can precipitate acute heart failure or be an alternative cause of patient's symptoms
- Elevated total white blood cell count can indicate ongoing infection causing decompensated heart failure

2. Creatinine and electrolytes

- Identify renal impairment
- Identify electrolyte disturbances that occur with diuretic use

3. HbA1c, lipid panel

- Identify risk factors for cardiovascular disease

4. Liver function test

- Elevated transaminases secondary to hepatic congestion
5. Thyroid function test
 - Especially in the presence of AF
 6. Iron panel
 - Iron deficiency due to both absolute and functionally depleted iron stores can occur with or without anaemia in heart failure

INVESTIGATIONS AT TERTIARY CARE

1. Echocardiography
2. Other investigations, e.g., plasma B-type natriuretic peptide, non-invasive imaging
3. Coronary angiogram - the current gold standard method for demonstrating coronary artery disease

MANAGEMENT AND FOLLOW-UP

Non-Pharmacological / Lifestyle

A. Management of Risk Factors

1. Lipid levels: Risk stratify and target as per latest ACG lipid 2023 (aim LDL < 1.4mmol/L in patients with history of ACS and LDL < 1.8mmol/L in stable IHD, post PCI or CABG)
2. Hypertension: Aim for BP < 130/80 mmHg
3. Diabetes mellitus: Screen for diabetes (Aim for HbA1c < 7% for diabetics)
4. Smoking: Offer smoking cessation clinic, counselling, behavioural advice
5. Weight reduction: Consider for patients with BMI ≥ 23g/m²
6. Alcohol: Advise all patients to abstain from alcohol

B. Psychosocial Management

1. Identify psychosocial problems, e.g., depression, anxiety, social isolation
 - SSRIs have safer cardiovascular profiles

C. Physical Activity and Exercise Training Advice

1. All patients with heart failure should be encouraged to enrol in a multidisciplinary care cardiac rehabilitation programme.
2. General advice for moderate intensity aerobic physical activity 30 minutes at least 5 times a week.
3. Heart failure patients with stable NYHA Class II to III heart failure, with no contraindications, are encouraged to undertake exercise. Exercise intensity and duration should be determined by a trained physician or physiotherapist.

D. Patient Education

1. General information, e.g., explanation of symptoms, treatment of heart failure, prognosis, outlook, stress management
2. Self-monitoring, e.g., home BP monitoring, daily weight, fluid restriction, salt restriction
3. Heart failure patients with erectile dysfunction can be treated with PDE-5 inhibitors in the absence of significant myocardial ischaemia or concomitant use of nitrates. With regards to sexual activity, patients are advised to:
 - a) choose a time when rested, relaxed and free from stress
 - b) avoid sex after a heavy meal
 - c) select a familiar setting

- d) stop and rest if uncomfortable or tired during coitus
4. Advice on travel: avoid high altitudes or very hot places. Short air flights preferred over long journeys
5. Advice on list of medications to be avoided and plan of diuretic adjustment for weight increase if necessary
6. Dietary recommendations: DASH diet, advise on fluid restriction, how to read food labels, salt restriction
7. Action plan: teach patients to recognise significance of symptoms and signs, e.g., weight gain
8. Advance care planning for patients with end stage heart failure, home hospice/care

Pharmacological / Medications

Medications Used in Heart Failure Patients When Discharged from Hospitals

Drug Class / Drug	Strength per Tablet	Initial Dose	Maximum Dose	Indications
Angiotensin receptor neprilysin inhibitors (ARNI) Sacubitril/Valsartan \$\$\$\$^ \$\$\$\$\$^ (For 200mg tablets)	24 mg/26 mg 49 mg/51 mg 97 mg/103 mg	49 mg/51 mg bd	97mg/103 mg bd	• First line therapy for patients with HFrEF to reduce morbidity and mortality • Recommended for 2DE to review EF prior to initiation • Usually started by Cardiologists • Avoid initiation if eGFR < 30ml/min. Once initiated, can continue even till dialysis. • May be associated with decreased renal function and increased risk of hyperkalaemia • Caution in patients with renal artery stenosis • Contraindicated in patients with history of angioedema with ACE inhibitors or ARBs, pregnancy or concomitant use of aliskiren-containing products • Recommended as a replacement for ACE inhibitors or ARB to further reduce the risk of HF hospitalisation and death ○ Requires a washout period of 36 hours when switching to and from ACEi to prevent angioedema ○ Routine combination with ACEi and ARB not recommended

Drug Class / Drug	Strength per Tablet	Initial Dose	Maximum Dose	Indications
ACE Inhibitors Captopril \$^ Enalapril \$^ Perindopril \$\$^ Lisinopril \$^	12.5, 25 mg 5, 10, 20 mg 4 mg 5, 10, 20 mg	6.25 mg tds 2.5 mg bd 2 mg om 2.5 – 5 mg om	50 mg tds 10 - 20 mg bd 8 - 16 mg om 20 - 40 mg om	- ACE inhibitors are beneficial to reduce morbidity and mortality in patients with HFrEF when the use of ARNI is not feasible - Dosages should be titrated upwards to dosages shown to be effective in controlled trials, e.g., captopril 50 mg tds and not titrated based on symptomatic improvement alone
Angiotensin Receptor Blockers (ARB) Losartan \$^ Valsartan \$^ Candesartan \$\$^	50 mg 80 mg 4 mg	25 - 50 mg om 20 - 40 mg bd 4 - 8 mg om	50 - 100 mg om 160 mg bd 32 mg om	- ARB should be used as an alternative therapy in patients who are ACE inhibitor intolerant and when the use of ARNI is not feasible - Routine use of combination of ACE inhibitor, ARNI and ARB is not recommended
Sodium-glucose co-transporter 2 (SGLT2) inhibitors Dapagliflozin \$\$\$^ Empagliflozin \$\$^	10 mg 10, 25 mg	10 mg om 10 mg om	10 mg om 10 mg om	- In both DM and non-DM patients, SGLT-2 inhibitors reduce HF hospitalisation and cardiovascular mortality in symptomatic HF (NYHA II-IV) irrespective of ejection fraction - In patients with renal impairment, initiation of dapagliflozin is not recommended at eGFR <25ml/min. However, in patients who have previously tolerated therapy, it can be continued at the same dose until initiation of dialysis. For empagliflozin, benefits have been shown in patients with eGFR ≥20ml/min.
Beta-Blockers Carvedilol \$\$^ Bisoprolol \$^	6.25, 25 mg 2.5, 5 mg	3.125 mg bd 1.25 mg om	25 mg bd (> 85 kg: 50 mg bd) 10 mg om	- Standard therapy for clinically stable patients with LV systolic dysfunction (EF ≤ 40%) and mild-moderate heart failure (NYHA class II - III) - Indicated in those who are stable and not in acute pulmonary oedema or severe heart failure, especially when EF ≤ 40%, NYHA class II – III - Only carvedilol, bisoprolol and sustained release metoprolol (succinate) (not available in NUP) have been shown to be effective in reducing risk of death and hospitalisation in heart failure. - Cautious dose titration every 2 - 4 weekly

Drug Class / Drug	Strength per Tablet	Initial Dose	Maximum Dose	Indications
				Contraindicated in hypotension, bronchospasm, pulmonary oedema, symptomatic bradycardia, 2nd, or 3rd degree heart block
Mineralocorticoid Receptor Antagonists (MRA) Spironolactone \$^A	25 mg	12.5 - 25 mg om	25 mg om or bd	- Used for symptomatic patients (NYHA classes II–IV) with HFrEF (EF ≤ 35%) already on an ACE inhibitor (or an ARB) and a beta-blocker, unless contraindicated - Recommended in patients following a myocardial infarction who develop HFrEF (EF ≤ 40%) or have a history of DM - Can result in breast discomfort and gynaecomastia in male patients, reversible and dose related. Can switch to eplerenone - Contraindicated if eGFR < 30 ml/min - Requires careful monitoring of renal function and serum K
Diuretics Hydrochlorothiazide \$^A Frusemide \$^A	25 mg 40 mg	12.5 mg om 20 mg om	50 mg om 160 - 200 mg (as a single dose)	- For all symptomatic patients to improve symptoms and relief congestion - Continued indefinitely even when rendered free from oedema. Once euvolaemia is attained, the diuretic dose should be decreased, to a minimum dose needed to maintain clinical euvolaemia. The dose should be regularly reassessed, as it may need to be adjusted according to volume status - Creatinine should be checked when adjusting the dose to avoid overdiuresis - HCTZ (avoid if eGFR < 30 - 40 ml/min) - for mild heart failure - Frusemide - for more severe heart failure - In severe heart failure, thiazides have synergistic effect with loop diuretics - Concurrent use with SGLT2-I may potentiate effects of diuretics
Digoxin \$^A	0.0625, 0.25 mg	0.0625 mg	0.5 mg	- Can be considered for patients with symptomatic heart failure (NYHA class II - IV) who are on standard therapy. - Does not improve long term survival but reduces rehospitalisation - Indicated for patients with AF and CHF (Beta-blockers are preferred) - Used as an add on therapy to beta-blockers - Contraindicated in patients with bradycardia, ventricular arrhythmia and severe renal dysfunction

Drug Class / Drug	Strength per Tablet	Initial Dose	Maximum Dose	Indications
Hydralazine + \$^	10, 25, 50 mg	10 mg 6H	225 - 300 mg/day in divided doses	- Usually indicated for those intolerant of ARNI / ACE inhibitors / ARBs - Add on therapy in symptomatic patients who are on optimal medical therapy - High incidence of side-effects such as headache
Isosorbide Dinitrate (ISDN) \$^ <i>(Not available in NUP)</i>	10 mg	20 mg 3 - 4 times/day	120 - 160 mg/day in divided doses	

^ Cost \$ <\$10 per month \$\$\$\$ \$30 - <\$60 per month
 \$ \$ \$10 - <\$20 per month \$\$\$\$\$ \$60 - <\$90 per month
 \$ \$ \$ \$20 - <\$30 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

4 Pillars of HFrEF Management

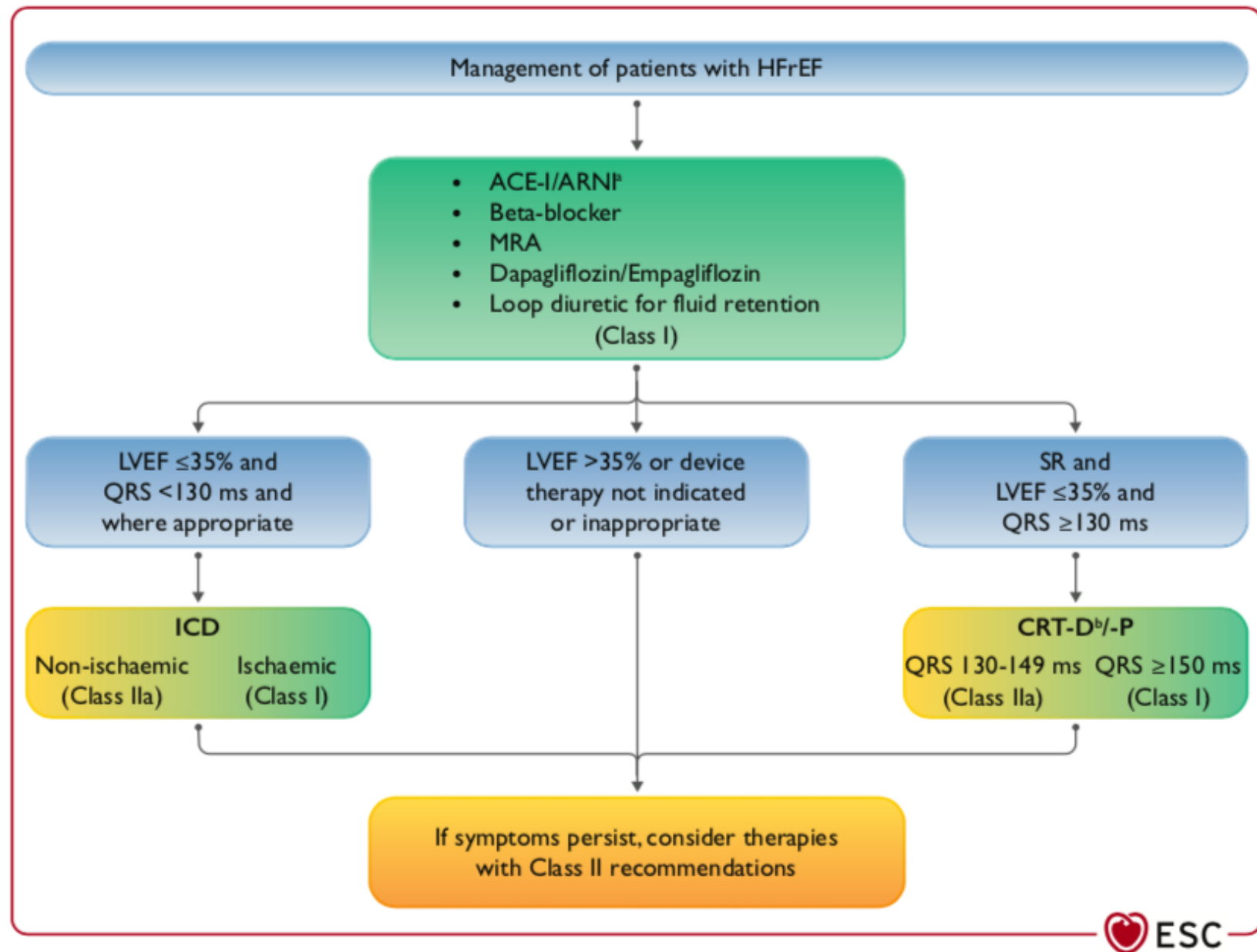
1. *Angiotensin receptor neprilysin inhibitors (ARNI) / Angiotensin receptor blockers (ARB) / Angiotensin-converting enzyme (ACE) inhibitors
2. Sodium-glucose co-transporter 2 (SGLT2) inhibitors
3. Beta blockers
4. Mineralocorticoid receptor antagonists (MRA)

Pharmacological Therapy for HFrEF and HFmEF

	For Survival / Morbidity	For Symptoms
Stage A (Patients at-risk)	Identify and control underlying cardiovascular risk factors e.g., Hypertension, Hyperlipidaemia, Diabetes mellitus SGLT2 inhibitors for patients with diabetes mellitus	No HF symptom.
Stage B (Pre-HF)	Stage A interventions + ACE inhibitor / ARB Beta-blocker <i>Class I indication for HFrEF</i> <i>Class IIb indication for HFmEF</i>	No HF symptom.
Stage C (Symptoms / signs present)	Stage A interventions + *ARNI / ACE inhibitor / ARB *Replace ACE inhibitor / ARB with ARNI if symptomatic Beta-blocker Mineralocorticoid antagonist SGLT2 inhibitor	Diuretics as needed. Refer to EMD / Cardiology for further management.
Stage D	Stage C interventions	Refer to EMD / Cardiology for further management.

Adapted from ACCF/AHA Heart failure guidelines (2022)

**Preferred option but recommended for 2DE to review EF prior to initiation. Usually started by Cardiologists.*



Management of HFpEF

Recent advances in HFpEF treatment have emerged in the past few years. SGLT2 inhibitors have been shown to significantly reduce the combined risk of cardiovascular death or hospitalisation for patients with HFpEF irrespective of diabetes status at baseline (*EMPEROR Preserved trial 2021 and DELIVER trial 2022*).

- There are few disease modifying therapies available in the treatment of HFpEF. The aims are to Identify and treat the underlying risk factors, aetiology, and co-existing comorbidities in HFpEF (e.g., hypertension, coronary artery disease, atrial fibrillation and valvular heart disease)
- Reduce symptoms of congestion with diuretics

Role of Health Team Members

1. Family Physician

- To provide management of HF, including health promotion and prevention / detection / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centered care

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient
- Introduce and enrol patient onto Teleconsult
- Perform general screening (i.e., fall risk, social economics, smoking & drinking history)
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza and pneumococcal) and cancer screenings (FIT kit, mammogram, PAP smear / HPV)
- Recruitment into PTEC – HT/DM

3. Care Manager

- Patient education on BP targets, LDL targets and glucose targets, and lifestyle measures
- Teach home blood pressure monitoring to patient and validate BP set
- Assess for treatment adherence

4. Advanced Practice Nurse (APN)

- Manage patients with diabetes, hypertension and dyslipidaemia within scope of practice
- Titrate medications for hypertension, diabetes and hyperlipidaemia
- Manage common acute conditions such as upper respiratory tract infection, gastroenteritis should the patient concomitantly present with these reasons for encounter

5. Clinical Pharmacist

- Manage patients with diabetes, hypertension and dyslipidaemia within scope of practice
- Titrate medications for hypertension, diabetes and hyperlipidaemia

- For patients who need drug optimisation or closer monitoring due to increased risk for drug interactions, polypharmacy, lack of drug-related knowledge or drug non-adherence

6. Dietitian

- Patient education on DASH diet
- Patient education on weight management
- Provide dietary / nutritional support / advise on food choices

7. Psychologist

- Psychological and behavioural interventions to manage psychological stress, improve disease management and quality of life.
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders etc.)
- Psychological support for carers experiencing caregiver stress

8. Medical Social Worker

- Conduct biopsychosocial assessment
- Provide intervention for social assistance including application of schemes and services, and referral for community resources and caregiver support to support with living with their chronic disease

9. Financial Counsellor

- Financial counselling advice with patients raising difficulties in managing their medical bills
- Advise on MediSave utilisation for chronic disease polyclinic bills
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high-cost drugs
- Application on Institutional Medical Fund-Remission (if applicable for non-Singaporeans)

10. Pharmacist

- Smoking cessation clinic
- Detect, prevent and address drug-drug / drug-disease interactions
- Perform medication reconciliation

SPECIAL SITUATIONS

Diagnosis and management of acute decompensated heart failure

1. Recognise symptoms and signs of decompensation
2. Consider precipitating causes
3. Investigations in primary care
4. Primary care management
 - If the underlying cause of decompensation can be confidently elicited; most common being lack of adherence to fluid restriction or medications, such patients can be treated in primary care and given a timely review
 - Ensure good control of cardiovascular risk factors
5. Considerations for Emergency Department referrals
 - Precipitating causes that need to be addressed: Hb drop, acute coronary syndromes, new onset arrhythmias e.g. atrial fibrillation, severe infections *etc.*

Urgent and Emergency Situations

1. A&E Referrals

- All patients with acute onset, moderate to severe heart failure including acute pulmonary oedema and cardiogenic shock
- Patients with recurrent heart failure complicated by acutely threatening events or clinical situations, e.g., recent AMI, pulmonary or systemic embolus, symptomatic arrhythmias

Fitness Certification

SMA guidelines 2011

- NYHA Class I and II: Annual review for fitness to drive
- NYHA Class III and IV: Permanently unfit

REFERRALS AND COMMUNITY RESOURCES

Referrals to Hospital

SOC Referrals

- All first-time heart failure patients
- Mild to moderate chronic heart failure (NYHA II to IV) with symptoms that are progressive and refractory to treatment
- Persistent symptoms of heart failure (chest pain, syncope, dyspnoea, or palpitations)
- Secondary causes, e.g., thyroid disease
- Pre-existing or developing metabolic abnormalities, e.g., sodium < 130 mmol/L and renal impairment, especially rising creatinine of at least 2-fold or to > 200 µmmol/L whichever is higher
- Abnormal weight loss due to malnutrition

HEART FAILURE SHARED CARE PROGRAMME

The Cardiology SOC may refer patients with stable heart failure on their follow-up to the Heart Failure Shared Care Programme which will alternate these visits between the Cardiology SOC and NUP.

Suitable patients for this programme include the following:

- **At least 6 months after last admission for acute decompensated heart failure (ADHF)**
- No need for active up-titration of disease-modifying medications
- Stable dose of diuretic medications
- On maximally tolerated dose of disease-modifying therapies HFrEF:
 - ARN-I/ ACE-I/ ARB (ARN-I preferred) + beta-blocker + MRA + SGLT-2 inhibitor
- HFmrEF/ HFpEF: SGLT-2 inhibitor

Escalation Criteria

1. Worsening heart failure
 - a) Worsening effort tolerance
 - b) Development of signs/ symptoms of volume overload
 - c) Recurrent outpatient visits for diuretic dose adjustment
2. Inability to tolerate HF therapy (e.g. new onset hypotension/bradycardia)
3. Syncope/ dizziness on HF therapy
4. New onset atrial fibrillation that is highly symptomatic or with poor rate control (<40bpm or >110bpm at rest)

Any of these patients who meet the escalation criteria above should be referred back to the discharging cardiologist by creating a Direct Access referral to NUHS Cardiology along with contacting the Community Cardiology Coordinator via:

- nuhcs_communitycardio@nuhs.edu.sg
or
- hotline at 8908 3194 (Mon – Fri, 9am to 5pm only)

Escalation back to SOC's only applies to haemodynamically stable patients.

Unstable patients should be admitted through Emergency Department

More information on the Heart Failure Share Care Programme can be found [here](#).

RECOMMENDED CARE COMPONENTS

Recommended Care Components	Minimum Frequency	Remarks
Blood Pressure Measurement	Twice a year	Aim BP < 130/80 mmHg
Weight and BMI Assessment	Twice a year	Keep < 23kg/m ² (for non-Asian population, keep BMI < 25 kg/m ²)
Lipid Profile	Annually	Risk stratify and target as per latest ACG lipid 2023 (aim LDL < 1.4mmol/L in patients with history of ACS and LDL < 1.8mmol/L in stable IHD, post PCI or CABG)
Smoking Assessment	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Offer smoking cessation clinic, counselling, behavioural advice
Diabetes Screening	Annually or once every three years, as clinically indicated	Screening should be carried out every three years for those with normal HbA1c or glucose tolerance, and annually for those with impaired fasting glycaemia (IFG) or impaired glucose tolerance (IGT)
Kidney Function Monitoring	Annually	More frequently if titrating HF medications or diuretics
Influenza Vaccination	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS)
Pneumococcal Vaccination	As recommended under NAIS	As recommended under NAIS
Shingles Vaccination (Recombinant herpes zoster vaccine)	2 doses at an interval of 2 to 6 months	As recommended under NAIS (patients aged 60 years or older)

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